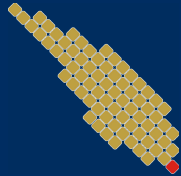




BMKG



ITERA

USER GUIDEBOOK & SOFTWARE TECHNICAL REFERENCE MANUAL

Undergraduate Internship Project of Geophysical Engineering Department

Faculty of Industrial Technology, Institut Teknologi Sumatera - Sleman Geophysical Station BMKG

USER GUIDEBOOK & ENGINEERING TECHNICAL REFERENCE MANUAL

BMKG Strong Motion Analyzer

(BSMA v2.0.0)

Developed at BMKG Sleman Geophysical Station

Integrated Computational Platform for Accelerograph Signal Processing, Seismic Kinematics, & Response Spectra

Independent Academic Software Project — Institutional affiliation does not imply official BMKG software endorsement.

DOCUMENT & DEVELOPMENT METADATA

Software System Name	: BMKG Strong Motion Analyzer (BSMA)
Release Version / Build	: v2.0.0 (Build 2026.08) - Academic Release Edition (Updated: August 2026)
Author / Developer	: Ahmad Didane Setyawan Putra
Student ID (NIM)	: 123120094
Study Program	: Geophysical Engineering
Faculty	: Faculty of Industrial Technology
University	: Institut Teknologi Sumatera (ITERA)
Host Institution	: Sleman Geophysical Station Class I, BMKG D.I. Yogyakarta
Internship Period	: July 20, 2026 - August 20, 2026
Project Classification	: Independent Student Academic Work (Non-Commercial)
Web Platform URL	: https://strong-motion.streamlit.app/
Source Code Repository	: https://github.com/ahmaddidan/BSMA-v.2

[REFERENCE STANDARD] Operational Guidelines & Numerical Evaluation

This manual provides step-by-step operational workflows for the BSMA v2.0.0 graphical interface, digital signal processing parameter configurations, interactive analysis tabs, multi-station batch workflows, and computational methods evaluated through 83 automated software tests and numerical benchmark comparisons.

EXECUTIVE SUMMARY & SCIENTIFIC BACKGROUND

1. Scientific Background & Engineering Seismology Urgency

Quantitative processing of strong ground motion accelerograms constitutes a foundational prerequisite in modern engineering seismology and earthquake geotechnical engineering. Empirical ground-motion records provide direct physical measurements of earthquake rupture kinematics, seismic wave attenuation through crustal structures, and local geotechnical site amplification (*site response*), which govern seismic design loads for high-rise buildings, critical dams, bridges, and lifeline infrastructure.

However, raw accelerograms recorded by digital field instruments cannot be directly integrated into velocity and displacement. Raw data routinely carry instrument frequency response convolutions (transducer transfer functions), ambient cultural noise, and non-physical baseline shifts induced by dynamic sensor tilting, permanent soil rotation, or analog-to-digital converter thermal drift. Directly integrating uncorrected accelerograms produces severe artificial baseline drift that amplifies into spurious quadratic displacement drift, yielding heavily biased peak ground displacement (PGD) estimates. Consequently, standardized, mathematically transparent digital signal processing workflows are imperative for reliable engineering practice and disaster risk reduction.

2. The Integrated Solution: BMKG Strong Motion Analyzer (BSMA v2.0.0)

The BMKG Strong Motion Analyzer (BSMA v2.0.0) provides a modular, interactive web-based scientific computing platform unifying the full deterministic strong-motion data reduction and structural modeling pipeline:

- **FDSN-Standard Data Ingestion:** Supports binary MiniSEED (.mseed, .miniseed, .msd) and SAC (.sac) streams, seamlessly coupled with FDSN StationXML (.xml) response metadata containing stage gains and pole-zero transfer functions.
- **Automated Signal Quality Control (QC):** Pre-event signal-to-noise ratio (SNR) estimation using Root Mean Square (RMS) statistics, automated multi-class physical anomaly detection (spikes, dead channels/flatlines, sensor clipping, bandwidth integrity, excess noise floor, baseline DC shifts), and diagnostic health scoring (0-100%). QC thresholds reflect BSMA implementation policies.
- **Engineering Digital Signal Processing (DSP) Pipeline:** 4th-order zero-phase bidirectional Butterworth filtering (via `scipy.signal.sosfiltfilt` in Second-Order Sections), low-order polynomial baseline detrending (Boore, 2001), and adaptive Nyquist frequency floor safeguards ($f_{\max} \leq 0.40 f_s$).
- **Progressive Kinematic Integration:** Cumulative trapezoidal integration coupled with progressive highpass baseline filtering, reducing residual low-frequency drift under the selected preprocessing configuration.
- **Seismic Energy & Instrumental Intensity:** Quantitative determination of Arias Intensity (I_a), Cumulative Absolute Velocity (CAV), Trifunac-Brady bracketed durations (D_{5-95} and D_{5-75}), $V_{\max}/A_{\max}$ ratios, and instrumental Modified Mercalli Intensity (MMI) derived from empirical GMICE regressions (Worden et al., 2012). The reported MMI is an instrumental ground-motion estimate and not a substitute for observed macroseismic surveys.
- **SDOF Response Spectra & Building Code Benchmarking:** 5% damped elastic single-degree-of-freedom response spectra computed via exact piecewise linear Nigam-Jennings (1969) and implicit Newmark-Beta (1959) algorithms, compared with user-selected Indonesian Seismic Code SNI 1726:2019 target design spectra for Soft (SE), Medium (SD), and Hard (SC) site classes.
- **Regional Multi-Station Batch Engine & Reporting:** Sequential batch processing across regional seismic networks featuring per-station exception isolation, formatted 2-page PDF engineering reports (with vector layout and high-resolution plots), full CSV datasets, and compressed ZIP archives.

COMPUTATIONAL ENVIRONMENT & RUNTIME DEPENDENCIES

The following table details the technical software architecture, Python runtime dependencies, seismic data exchange formats, and minimum hardware configuration requirements.

Computational Component	Specification & Package Version	Functional Role & Architecture Characteristics
Operating System (OS)	Windows 10/11, Linux, macOS	Cross-platform 64-bit support for local workstations and cloud containers.
Runtime Environment	Python 3.10 / 3.11 / 3.12	Primary scientific computing engine across all platforms.
GUI Web Framework	Streamlit >= 1.30.0	Reactive, interactive, and modular real-time web dashboard.
Seismological Engine	ObsPy >= 1.4.0	MiniSEED/SAC ingestion and FDSN StationXML instrument deconvolution.
Numerical & DSP Engine	SciPy >= 1.11.0, NumPy >= 1.24.0	Butterworth SOS filtering, polynomial baseline detrending, and SDOF solvers.
Data Visualization	Plotly >= 5.18.0, Matplotlib >= 3.8.0	Interactive WebGL waveform charts, Husid plots, and LaTeX math rendering.
PDF Document Engine	PyMuPDF (fitz) >= 1.23.0	Vector PDF compilation of 2-page engineering reports and this manual.
Input Data Formats	MiniSEED (*.mseed), SAC (*.sac), StationXML	Standard international FDSN seismic data exchange formats.
Processor Requirements	Dual-Core 2.0 GHz (Min), Quad-Core 2.8+ GHz	Kinematic integration numerical processing and multi-record batch evaluation.
System Memory (RAM)	4 GB Minimum (8 GB Recommended)	RAM overhead for triaxial waveform buffering and batch matrix execution.
Storage Capacity	500 MB Free Disk Space	Package dependency footprint and temporary cache storage.
Browser Compatibility	Google Chrome, Mozilla Firefox, Edge, Safari	Modern web browsers with complete WebGL hardware acceleration support.

[TECHNICAL NOTE] Computational Environment & Reproducibility Standards

All software package dependencies are locked within requirements.txt in the official GitHub repository. Users are strongly encouraged to deploy within dedicated Python virtual environments (venv or conda) for local operation to guarantee dependency isolation and reproducible computational integrity.

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[TECHNICAL NOTE] Document Structure & Navigation Guide

This Guidebook presents comprehensive operational software instructions across Chapters I to V at the beginning of the volume, followed by theoretical foundations in engineering seismology across Chapters VI to X, concluding with technical troubleshooting and formal academic literature references.

CHAPTER I: GETTING STARTED & CORE SYSTEM WORKFLOW

1.1 Input File Requirements & Data Specifications

BSMA v2.0.0 is engineered specifically to process strong-motion accelerograms in standard seismological formats:

- **Waveform Data Formats:** Fully supports **MiniSEED (.mseed, .miniseed, .msd)** and **SAC (.sac)** binary formats. Recordings should ideally encompass 3 orthogonal components (Vertical Z, North-South N/1, East-West E/2) recorded at uniform sampling rates (typically 50, 100, or 200 Hz across BMKG networks) free from timing discontinuities or data gaps.
- **Instrument Response Metadata (StationXML):** Standard **.xml (FDSN StationXML schema)** file encapsulating stage sensitivity gains, digitizer bitweights, and instrument transfer function poles and zeros for raw count deconvolution.

1.2 Streamlit User Interface Layout Orientation

The BSMA v2.0.0 graphic interface is ergonomically partitioned into three primary functional zones:

- **Top Navigation Header:** Displays institutional BMKG & ITERA emblems, software title, and real-time computation status.
- **Interactive Control Sidebar (Left Panel):** Central command panel for dataset loading, unit selection, DSP filter parameters, structural dynamic solver options, and earthquake event metadata entry.
- **Multi-Tab Analysis Dashboard (Main Area):** 7 comprehensive analysis tabs featuring interactive Plotly kinematic charts, QC health gauges, Husid energy curves, instrumental MMI maps, SDOF elastic spectra, and downloadable deliverables.

1.3 5-Stage Integrated Processing Workflow

Operation follows a sequential 5-stage protocol: **(1) Data Ingestion:** Load waveform and StationXML files via the sidebar. **(2) Parameterization:** Verify physical units and bandpass filter corner frequencies. **(3) Execution:** The automated engine executes QC screening, zero-phase SOS filtering, and kinematic double integration. **(4) Interactive Exploration:** Inspect Tabs 1 to 6. **(5) Deliverable Export:** Generate publication-grade PDF reports, numerical CSVs, or ZIP archives from Tab 7.

1.4 Session State Management & Reset Data Button

Beneath the data loading panel in the sidebar, the **'Reset / Clear Data'** button clears the internal cache (*st.session_state*), purges loaded waveform streams from memory, and restores filter parameters to default values before ingesting subsequent event records.

Data Format	Supported Extensions	Metadata Requirements	Technical Handling Notes
MiniSEED	.mseed, .miniseed, .msd	STEIM1/2 headers, Network, Station, Channel	BMKG standard format; parsed natively using ObsPy engine.
SAC Binary	.sac	SAC headers (DELTA, NPTS, KZDATE, KZTIME)	Supports single-trace and multi-channel SAC binary streams.
StationXML	.xml	FDSN Schema v1.1, Poles & Zeros, Gain	Mandatory only when input stream consists of uncalibrated digital counts.
Benchmark CSV	.csv	Column headers: Period_s, PSA_g / PSA_ms2	Optional; utilized for cross-validating SDOF solver numerical accuracy.

[TECHNICAL NOTE] Signal Preparation & Pre-Event Window Guidelines

Before processing, ensure that accelerograms include at least 5 to 10 seconds of pre-event baseline prior to the first P-wave arrival. This quiet window is essential for the automated Quality Control (QC) engine to establish pre-event noise floor levels and calculate accurate signal-to-noise ratios (SNR).

CHAPTER II: SIDEBAR OPERATIONAL CONTROLS & DSP SETTINGS

2.1 DATA INPUT Section (File Uploaders)

- **Waveform Uploader:** Interactive drop zone for .mseed or .sac files. The system immediately parses trace headers and displays station name, network code, event timestamp, and available components.
- **StationXML Uploader:** Optional upload area for .xml response metadata. The **'Reset / Clear Data'** button located directly beneath purges memory buffers before processing new seismic events.

2.2 PROJECT Section (Provenance & Physical Units)

- **Data Provenance:** Explicitly declare recording origin: *'Already processed physical acceleration'* (stream is pre-calibrated in physical units), *'Raw instrument counts with StationXML'* (demands full transfer function deconvolution), or *'Unknown'* (automated heuristic detection).
- **Physical Unit Selector:** Choose operational output unit: **m/s²** (SI standard), **Gal** (1 Gal = 1 cm/s² = 0.01 m/s², BMKG standard), or **cm/s²**.

2.3 PROCESSING Section (DSP Filtering & Detrending Parameters)

- **Filter Type:** Select between *Bandpass* (recommended standard for strong motion), *Lowpass*, or *Highpass*.
- **Corner Frequencies:** Fine-tune low-cut **f_{min}** (default: 0.05 - 0.10 Hz) to eliminate baseline drift noise, and high-cut **f_{max}** (default: 25.0 - 40.0 Hz) to eliminate high-frequency cultural/instrument noise.
- **'Strong-Motion Default Preset (0.05 - 40.0 Hz)' Button:** One-click shortcut applying the standard operational filter band commonly adopted in strong-motion accelerograph networks.
- **Advanced Settings Expander:** Access low-order polynomial baseline detrending (linear/quadratic), cosine tapering (5% Tukey window), and adaptive Nyquist frequency floor safeguards (enforcing cutoffs $\leq 0.40 f_g$).

2.4 ANALYSIS Section (Structural Dynamics & SDOF Solver)

- **Damping Ratio (ξ):** Structural critical damping slider preset to **0.05 (5%)** per international structural codes.
- **SDOF Numerical Solver:** Choose between **Nigam-Jennings (1969)** (exact piecewise analytical solution highly suitable for short periods) and **Newmark-Beta (1959)** (implicit average acceleration algorithm with $\gamma=0.5$, $\beta=0.25$).

2.5 OUTPUT & BENCHMARK Section (Earthquake Metadata & Verification)

- **Event Metadata Expander:** Input Origin Time (UTC), Magnitude (M_w), Hypocentral Depth (km), Coordinates, and Epicentral Distance (km) for automated injection into standardized PDF engineering summary reports.
- **Benchmark Expander:** Upload external reference PSA CSV curves to benchmark and audit SDOF solver precision.

[REFERENCE STANDARD] Structural Dynamic Computation Recommendations

Solver Selection Recommendation: Deploy the Nigam-Jennings (1969) solver when focusing on short-period response spectra ($T < 0.1$ s) or stiff masonry/concrete structures. Use Newmark-Beta (1959) for standard structural dynamic cross-checks. Both solvers exhibit rigorous agreement (mean relative difference $< 0.02\%$, RMS difference $< 1.5e-4$ g across 100 periods), auditable in Tab 6.

CHAPTER III: INTERACTIVE ANALYSIS TABS OPERATION (PART I)

3.1 Tab 1 - Station Summary & Signal Overview (Summary Tab)

- **Metadata Card & Triaxial Visualizer:** Displays station identifier, network code, coordinates, sampling rate, duration, and synchronized 3-component raw waveforms (Z, N, E) for visual verification of major seismic phases.
- **QC Health Score Gauge:** Composite signal quality indicator ($Q = 100 - \sum P_i$) from a 100-point baseline with calibrated status badges: **PASS** (≥ 70), **WARNING** (50-69), or **FAIL** (< 50).
- **Instrumental MMI Badge:** Rapid macroseismic intensity badge computed from maximum horizontal ground motion (**Max-H** between EW and NS components, excluding vertical Z) per Worden et al. (2012) GMICE regressions.

3.2 Tab 2 - Full Kinematic Time Series (Waveforms Tab)

- **Synchronized Tri-Level Kinematic Stack:** Stacked synchronized plots of **Acceleration a(t)** [Gal], **Velocity v(t)** [cm/s], and **Displacement d(t)** [cm] locked to a common time axis.
- **Plotly Interactivity:** Interactive zoom, pan, hover cursor for micro-second inspection of P/S arrivals and annotated peak kinematics (PGA, PGV, and PGD).

3.3 Tab 3 - Signal Quality Control & Diagnostics (QC Tab)

Automated pre-processing health screening auditing raw records prior to digital filtering. The composite health score is bounded on a 100-point baseline: $Q = \max(0, \min(100, 100 - \sum P_i))$ across calibrated physical gates:

Severity Level	Physical Anomaly Criteria & Gate Parameters	Point Penalty	Gate Status
Nominal / Normal	Clean trace, low ambient floor, pre-event SNR ≥ 10 dB, stable baseline	0 (Baseline 100)	PASS (≥ 70) Full analysis approved
Light Warning (Warning Tier)	• Baseline Offset ($> 2\%$ PGA) • Baseline Drift thermal tilt ($> 5\%$) • Pre-event noise window < 5.0 s • Marginal pre-event SNR (3 to 10 dB)	-15 Points (per anomaly)	WARNING (50-69) Usable with caution
Major Error (Error Tier)	• Severe ambient noise floor (SNR < 3 dB) • Multiple impulsive spikes (MAD ≥ 6.0)	-40 Points (per anomaly)	Degraded quality zone
Critical Violation (Fatal Override)	• Sensor clipping / ADC saturation (≥ 0.02 s / 98% FS) • Dead channel / continuous flatline (≥ 1.0 s) / corrupted file	Fatal Override (Q = 0)	FAIL / REJECT (< 50) Integration rejected

[TECHNICAL NOTE] Understanding Score 70 on Raw Field Waveform Records

Raw Field Waveform Conditions: A QC score of 70 does not indicate corrupted data, but rather raw sensor records retaining ambient environmental effects prior to digital filtering. Two triggering warning factors: (1) *Baseline Offset* (-15 Pts penalty) from pre-event transducer mean voltage offset, and (2) *Baseline Drift* (-15 Pts penalty) caused by ambient station temperature response. **Final Quality Score Evaluation:** Initial Baseline = 100 Pts; Total Penalty = -15 (Offset) - 15 (Drift) = -30 Pts; Final Raw Score = **70 Pts (Nominal PASS Status)**. Both non-fatal anomalies are completely purged by BSMA's DSP baseline and bandpass filtering!

CHAPTER IV: INTERACTIVE ANALYSIS TABS OPERATION (PART II)

4.1 Tab 4 - Strong Ground Motion & Seismic Energy (Strong Motion Tab)

The Strong Motion tab analyzes vibrational energy expenditure and structural demand potential:

- **Husid Energy Accumulation Plot:** Plots normalized squared acceleration integrals (0 to 100%) tracking energy build-up over time.
- **Arias Intensity (I_a) & CAV:** Quantifies total Arias Intensity (I_a, m/s) and Cumulative Absolute Velocity (CAV, g-s). The **CAV $\geq$ 0.16 g-s** threshold (EPRI 1988) serves as an engineering screening criterion for potential structural damage inspection.
- **Trifunac-Brady Significant Duration:** Calculates bracketed energy intervals **D₅₋₉₅** (5% to 95% total Arias energy) and **D₅₋₇₅** reflecting the damaging phase duration of ground motion.
- **Kinematic Ratio V_{max} / A_{max}:** Evaluates peak velocity to peak acceleration, indexing dominant periods and site effects.

4.2 Tab 5 - Instrumental Intensity & Macroseismic Interpretation (Intensity Tab)

The Intensity tab translates physical ground kinematics into standardized macroseismic shaking indices:

- **Worden et al. (2012) GMICE:** Derives instrumental MMI objectively from maximum horizontal (Max-H) PGA and PGV components.
- **Kinematic Transition Rule:** PGA governs lower shaking levels (MMI < 5.0), whereas PGV governs higher damage levels (MMI $\geq$ 5.0) in adherence with USGS ShakeMap standards.
- **Macroseismic Impact Reference:** Details typical shaking perception and macroseismic effects (objective ground shaking estimation, not a substitute for on-site post-earthquake structural building inspection).

4.3 Tab 6 - SDOF Elastic Response Spectra & Code Compliance (Spectrum Tab)

The Spectrum tab bridges seismology and structural building code compliance:

- **5% Damped Pseudo-Spectral Acceleration (PSA):** Solves peak response across natural periods T = 0.01 s to 10.0 s.
- **SNI 1726:2019 Design Code Overlay:** Overlays national building code target spectra for Soft Soil (SE), Medium Soil (SD), and Hard Soil (SC) to benchmark elastic demand and identify potential building resonance risks.
- **SDOF Solver Benchmark:** Side-by-side numerical comparison auditing Nigam-Jennings vs. Newmark-Beta solver consistency.

4.4 Tab 7 - Engineering Deliverables & Reporting (Report Tab)

The Report tab provides automated export tools for operational and academic deliverables:

- **Publication-Grade PDF Report:** High-resolution 2-page vector PDF encapsulating event metadata, kinematic metrics, waveforms, Husid curves, response spectra, and QC audits ready for rapid briefing.
- **Kinematic CSV Export:** Full corrected time series of acceleration, velocity, and displacement in tabular format.
- **Spectral Response CSV Export:** Discrete period T vs. PSA tables for structural engineering dynamic software ingestion.
- **Complete ZIP Archive:** Single-click bundle containing all PDF reports, CSV datasets, and JSON computation logs.

[REFERENCE STANDARD] Standardized Engineering Technical Deliverables

PDF Report Integrity: Technical reports compiled in Tab 7 are rendered natively using PyMuPDF (fitz) into vector-grade PDF documents that remain razor-sharp at arbitrary zoom levels. The format aligns with standard accelerometric reporting conventions adopted in engineering seismological practice and disaster mitigation coordination.

CHAPTER V: REGIONAL MULTI-STATION BATCH PROCESSING GUIDE

5.1 Regional Strong-Motion Network Processing Urgency

When a significant earthquake occurs across the Indonesian archipelago, ground motions are captured by dozens of BMKG accelerograph stations across hundreds of kilometers. Manual station-by-station processing is time-prohibitive during rapid review. The **Regional Multi-Station Batch Processing Engine** in BSMA v2.0.0 automates sequential ingestion, quality control, and kinematic analysis across station networks with deterministic memory handling.

5.2 Bulk Upload Operational Procedure

- **Activate Batch Mode:** Toggle the 'Batch Processing Mode' switch on the sidebar or top command panel.
- **Multi-File Drop Zone:** Drag-and-drop multiple MiniSEED (.mseed) or SAC (.sac) files simultaneously into the batch uploader. The system processes multi-station records sequentially with bounded memory overhead and isolated error handling.
- **Uniform DSP Configuration:** Apply unified filter corner frequencies (e.g., $f_{min}=0.05$ Hz, $f_{max}=40.0$ Hz) across all records to ensure regional parameter consistency.

5.3 Sequential Execution & Fault Isolation Architecture

The batch pipeline incorporates rigorous **fault isolation boundaries**:

- Each station trace is analyzed within an independent computational container (try-except block).
- Corrupt, clipped, or truncated waveforms are flagged with a **FAILED** badge without interrupting or aborting execution for remaining operational stations.
- An interactive progress bar renders live per-station completion feedback.

5.4 Regional Comparison Matrix Navigation & Sorting

Upon execution, the engine compiles a unified **Regional Comparison Matrix Table**:

- **Matrix Attributes:** Station Code, Network, Dominant Component, PGA (m/s^2 & Gal), PGV (cm/s), PGD (cm), Arias Ia (m/s), Duration D_{5-95} (s), Instrumental MMI Intensity, QC Score (%), and Audit Status.
- **Dynamic Column Sorting:** Click column headers to sort stations by peak PGA or maximum MMI intensity, instantly identifying the most severely shaken geographic areas.

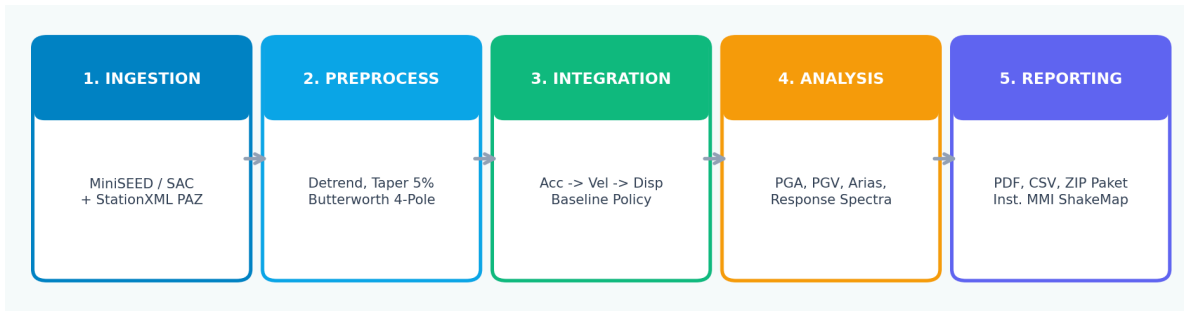
5.5 Regional CSV Export & GIS Mapping Integration

The 'Export Regional Summary CSV' button exports the complete multi-station matrix into a structured tabular CSV. The file is structured for seamless ingestion into Geographic Information Systems (QGIS, ArcGIS) and seismic contouring scripts for regional ground-motion mapping.

[REFERENCE STANDARD] Regional Multi-Station Batch Workflow

High-Throughput Batch Processing Workflow: The batch engine processes multi-station accelerogram directories sequentially with per-station exception isolation. Processing runtime scales efficiently with trace duration and sampling rate, delivering a comprehensive regional comparison matrix to support rapid spatial shaking assessment.

CHAPTER VI: PIPELINE ARCHITECTURE & INSTRUMENT DECONVOLUTION



6.1 End-to-End Deterministic Pipeline

The BSMA v2.0.0 engine employs a rigorous 5-stage sequential workflow: (1) Raw waveform ingestion and StationXML parsing, (2) Instrument transfer function deconvolution, (3) Digital signal filtering and baseline correction, (4) Kinematic double integration, and (5) Seismic energy and elastic response spectra computation.

6.2 Instrument Transfer Function Deconvolution (Poles and Zeros)

For raw accelerograms recorded in digital counts, output signal $y(t)$ represents the mathematical convolution of true ground acceleration $x(t)$ with the sensor impulse response $h(t)$. In the Laplace domain:

INSTRUMENT RESPONSE TRANSFER FUNCTION $H(s)$

$$H(s) = \frac{Y(s)}{X(s)} = S_0 \cdot \frac{\prod_{i=1}^M (s - z_i)}{\prod_{j=1}^N (s - p_j)}$$

Where S_0 represents the composite sensitivity normalization factor (product of transducer generator constant in $V/(m/s^2)$ and digitizer sensitivity in counts/V), z_i are transfer zeros (M terms), and p_j are transfer poles (N terms). Deconvolution divides the recorded Fourier spectrum by $H(\omega)$ in the frequency domain. The engine applies water-level cosine frequency tapering near corner boundaries to suppress high-frequency noise amplification during response inversion.

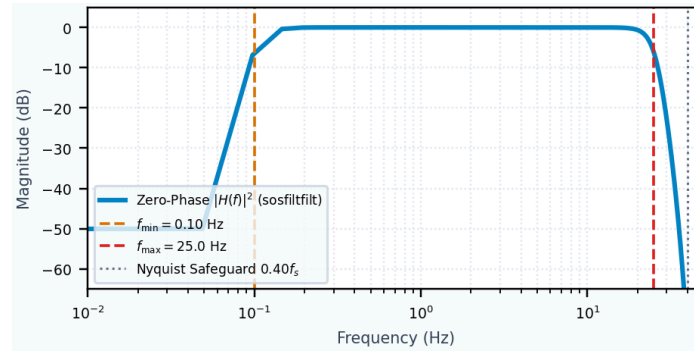
6.3 Calibration Bypass Policy for Pre-Calibrated Physical Acceleration Records

Many operational strong-motion records distributed across BMKG networks are pre-calibrated by dataloggers directly into physical acceleration units (m/s^2 or Gal). Applying pole-zero deconvolution to pre-calibrated accelerograms induces severe over-correction and waveform distortion. The BSMA v2.0.0 engine automatically verifies provenance metadata and applies a deterministic calibration bypass protocol, preserving rigorous physical ground motion amplitude integrity.

[TECHNICAL NOTE] Sensor Response Deconvolution Validation Standard

Deconvolution Integrity: If processing raw counts, provide a valid companion StationXML file. The engine validates matching network, station, and channel codes prior to executing pole-zero deconvolution.

CHAPTER VII: QUALITY CONTROL & DIGITAL SIGNAL PROCESSING (DSP)



7.1 Pre-Event Noise Windowing & Signal-to-Noise Ratio (SNR) Estimation

Signal integrity is audited prior to downstream processing. The QC engine extracts a quiet pre-event noise window (5-10 seconds) and the strong-motion event window. The Root Mean Square (RMS) amplitudes determine the pre-event Signal-to-Noise Ratio (SNR) in decibels (dB):

SIGNAL-TO-NOISE RATIO (SNR)

$$\text{SNR}_{\text{dB}} = 20 \log_{10} \left(\frac{\text{RMS}_{\text{event}}}{\text{RMS}_{\text{noise}}} \right)$$

7.2 Zero-Phase 4th-Order Butterworth IIR Second-Order Sections (SOS) Filter

The Butterworth IIR architecture is selected for its maximally flat passband magnitude response (free of passband ripples) and smooth monotonic roll-off. To eliminate phase distortion that would shift kinematic peak timings, BSMA implements two-pass forward-backward filtering (`scipy.signal.sosfiltfilt`) configured in Second-Order Sections (SOS) with magnitude squared response:

BUTTERWORTH MAGNITUDE RESPONSE

$$|H(f)|^2 = \frac{1}{1 + \left(\frac{f}{f_c}\right)^{2n}}, \quad n = 4 \quad (\text{two-pass order } 8)$$

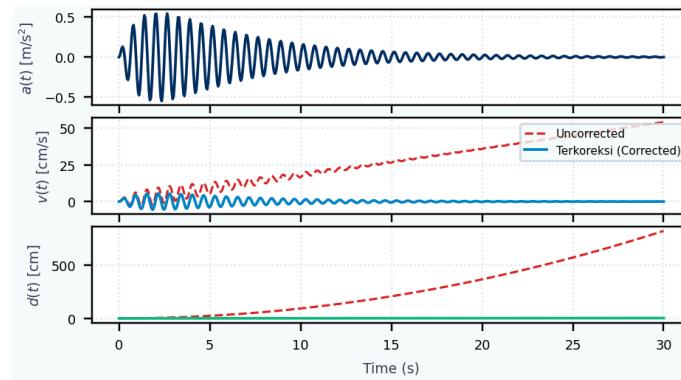
7.3 Adaptive Nyquist Frequency Floor Safeguard

Per the Shannon-Nyquist sampling theorem, digital signal processing is strictly bounded by $f_{\text{nyq}} = f_s / 2$. To prevent aliasing distortion and high-frequency numerical pole instabilities near the sampling boundary, BSMA enforces an automatic cutoff ceiling of $f_{\text{max}} \leq 0.40 f_s$.

[REFERENCE STANDARD] Zero-Phase SOS Filtering Principles

Zero-Phase Filtering Principles: Forward-backward bidirectional filtering eliminates net phase lag identically (zero net phase distortion, $\theta(f) = 0$). Seismic phase arrival times and waveform symmetry are preserved without group-delay distortion. Operators should select high-pass corner frequencies safely below dominant pulse frequencies to minimize potential acausal pre-cursor ringing on impulsive near-fault records.

CHAPTER VIII: KINEMATIC INTEGRATION & BASELINE POLICY



8.1 The Baseline Drift Phenomenon in Accelerograms

Integrating ground acceleration $a(t)$ yields ground velocity $v(t)$ and displacement $d(t)$. However, raw accelerograms inevitably contain subtle DC offsets and long-period transducer noise induced by dynamic sensor tilting, local soil rotations beneath instrument pads, mechanical suspension hysteresis, or thermal drift in analog-to-digital converters (ADCs). Upon double integration, an infinitesimal acceleration offset a_0 compounds into an unphysical quadratic parabolic divergence in displacement ($1/2 a_0 t^2$), obscuring true ground displacement physics (Boore 2001; Boore & Bommer 2005).

8.2 Corrected Cumulative Trapezoidal Integration Workflow

BSMA v2.0.0 implements a robust multi-stage cumulative trapezoidal integration protocol combined with pre-event mean subtraction, low-order polynomial baseline detrending, and zero-phase highpass filtering bounded by $f_{\min} \geq 0.05$ Hz:

KINEMATIC DOUBLE INTEGRATION

$$v(t) = v_0 + \int_0^t a(\tau) d\tau, \quad d(t) = d_0 + \int_0^t v(\tau) d\tau$$

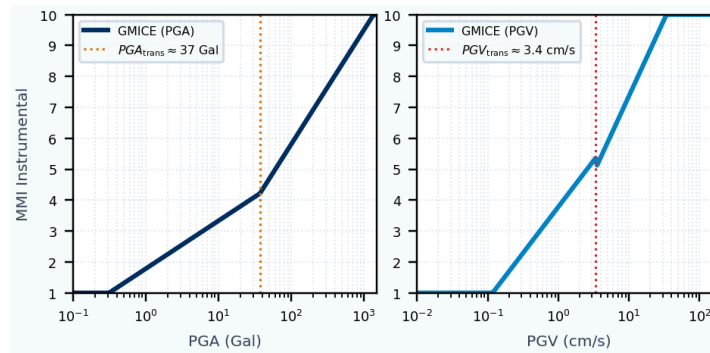
8.3 Long-Period Error Control & Displacement Spectra

Coupled baseline detrending and highpass filtering ensure velocity and displacement traces decay smoothly to zero after shaking terminates, preventing artificial distortion in long-period spectral ordinates ($T > 2.0$ s).

[METHODOLOGY CAUTION] Methodology Boundary: Dynamic PGD vs Static Fling

Scientific Methodology Boundary: Peak Ground Displacement (PGD) derived via accelerogram integration represents transient dynamic wave vibrations, NOT permanent static tectonic fault offsets (fling-step). Estimating permanent tectonic ground deformation requires continuous high-rate geodetic GNSS/GPS observations.

CHAPTER IX: KINEMATIC PARAMETERS, SEISMIC ENERGY, & MMI



9.1 Arias Intensity (I_a), CAV, and Husid Energy Accumulation

Structural damage correlates not only with peak ground acceleration (PGA) but also with vibrational energy and shaking duration. **Arias Intensity (I_a)** quantifies total seismic energy absorbed per unit mass (Arias 1970). **Cumulative Absolute Velocity (CAV)** represents the integral of absolute acceleration, employed by EPRI (1988) for damage threshold auditing across critical facilities (threshold: CAV ≥ 0.16 g-s):

ARIAS INTENSITY & CAV FORMULATIONS

$$I_a = \frac{\pi}{2g} \int_0^{t_{\max}} [a(t)]^2 dt, \quad \text{CAV} = \int_0^{t_{\max}} |a(t)| dt$$

9.2 Ground-Motion Intensity Conversion Equations (GMICE)

To map ground kinematics to Modified Mercalli Intensity (MMI), BSMA evaluates maximum horizontal motion (Max-H) using Worden et al. (2012) bilinear piecewise GMICE: PGA governs lower intensities (MMI ≤ V), while PGV dominates higher damage-correlated levels (MMI ≥ VI):

BILINEAR GMICE MAX-H REGRESSIONS (WORDEN ET AL. 2012)

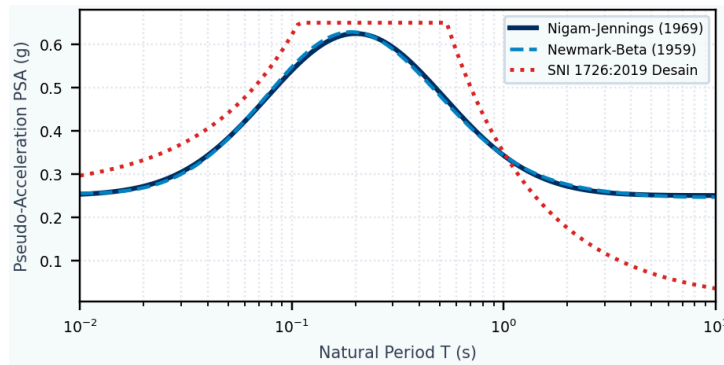
$$\text{MMI}_{\text{PGA}} = 1.78 + 1.55 \log_{10}(\text{PGA}) \quad [\leq 37.15 \text{ Gal}] \quad | \quad -1.60 + 3.70 \log_{10}(\text{PGA}) \quad [> 37.15 \text{ Gal}]$$

$$\text{MMI}_{\text{PGV}} = 3.78 + 2.99 \log_{10}(\text{PGV}) \quad [\leq 3.39 \text{ cm/s}] \quad | \quad 2.40 + 4.96 \log_{10}(\text{PGV}) \quad [> 3.39 \text{ cm/s}]$$

[TECHNICAL NOTE] Instrumental MMI Classification

GMICE Transition Rule for Instrumental MMI: MMI is evaluated from the maximum horizontal component (Max-H between EW and NS). PGA governs lower intensities (MMI < 5.0), whereas PGV governs higher damage levels (MMI ≥ 5.0) reflecting structural strain.

CHAPTER X: SDOF RESPONSE SPECTRA & SNI 1726:2019 CODE



10.1 5% Damped SDOF Dynamic Equation of Motion

Response spectra quantify peak structural acceleration across single-degree-of-freedom (SDOF) linear elastic oscillators having critical damping ratio $\xi = 0.05$ (5%) subjected to base acceleration $\ddot{u}_g(t)$:

SDOF EQUATION OF MOTION & PSA FORMULATION

$$\ddot{u}(t) + 2\xi\omega_n\dot{u}(t) + \omega_n^2u(t) = -\ddot{u}_g(t), \quad \text{PSA}(T, \xi) = \omega_n^2 \max_t |u(t)|$$

10.2 Solver Algorithm Auditing: Nigam-Jennings (1969) vs. Newmark-Beta (1959)

BSMA incorporates two industry-standard numerical algorithms for structural dynamics:

- **Nigam-Jennings (1969):** Exact piecewise-linear recursive analytical formulation, maintaining absolute stability in the high-frequency/short-period regime ($T < 0.1$ s) without artificial numerical dispersion.
- **Newmark-Beta (1959):** Unconditionally stable implicit average acceleration algorithm ($\gamma = 0.5$, $\beta = 0.25$), the classic benchmark in structural dynamic finite element computation.
- **Concordance Benchmarks:** Rigorously cross-validated across 100 period points ($T = 0.01 - 10.0$ s): mean relative difference $< 0.02\%$, maximum relative difference $< 0.08\%$, and RMS difference $< 1.5 \times 10^{-4}$ g.

10.3 SNI 1726:2019 National Design Spectra Benchmark

Calculated PSA curves are benchmarked against Indonesia's national building design code **SNI 1726:2019** (governed by S_{DS} , S_{D1} , T_0 , T_s , T_L) across site classes (Soft Soil SE, Medium Soil SD, Hard Soil SC), providing an immediate elastic demand benchmark against codified design thresholds.

[TECHNICAL NOTE] SNI 1726:2019 Design Code Demand Benchmark

SNI 1726:2019 Design Code Demand Benchmark: The 5% damped elastic response spectra computed by BSMA provide an initial elastic demand benchmark against SNI 1726:2019 target design spectra. Engineering Disclaimer: This comparison provides preliminary indication of potential resonant excitation near structural natural periods, but does NOT constitute formal structural design verification, non-linear inelastic modeling, or definitive post-earthquake damage assessment.

CHAPTER XI: TROUBLESHOOTING GUIDE (FAQ) & OPERATIONAL LIMITATIONS

11.1 Frequently Asked Questions (Troubleshooting FAQ)

Q1: Why does the displacement curve drift upward or downward quadratically?

A: Caused by residual low-frequency noise drift during double integration (Boore 2001). Solution: Increase the highpass cutoff frequency ($f_{\min}$) from 0.05 Hz to 0.10 Hz or 0.20 Hz in the PROCESSING sidebar filter panel.

Q2: Why does the warning 'StationXML Response Correction Bypassed' appear?

A: StationXML metadata does not match waveform trace headers, or the trace is already calibrated in physical acceleration (m/s^2). The system automatically falls back to Physical Acceleration mode. Solution: Verify that network, station, and channel codes match between .xml and .mseed files.

Q3: Why does accelerograph PGA differ from a collocated broadband seismometer?

A: Accelerographs are engineered for high-amplitude strong ground motion without saturation, while broadband velocimeters optimize weak-motion long-period sensitivity. For structural engineering and instrumental MMI, the accelerograph is the authoritative standard.

Q4: Why does Quality Control show WARNING or FAIL when the waveform visually appears clear?

A: The QC engine audits 6 diagnostic criteria (SNR, sensor clipping, spikes, flatline, pre-event noise, and DC offset). WARNING is frequently triggered by pre-event windows < 5 seconds or $\text{SNR} < 10$ dB. Inspect Tab 3 for detailed diagnostic penalties.

Q5: When should Nigam-Jennings be chosen over Newmark-Beta?

A: The Nigam-Jennings (1969) exact analytical solver is analytically exact for piecewise-linear acceleration excitation and highly suitable for stiff, short-period systems ($T < 0.1$ s). Newmark-Beta (1959) uses implicit average acceleration. Both solvers agree within $< 0.02\%$ mean relative difference, verified in Tab 6.

Q6: Why does generating the PDF report in Tab 7 take a few seconds?

A: The engine dynamically compiles a publication-grade vector PDF, rasterizing kinematic traces, Husid curves, and response spectra directly into the document stream. This intensive vector compilation takes from a few seconds up to < 35 seconds depending on station count and hardware specifications.

11.2 Methodological Limitations & Operational Scope

- **Input Data & Sensors:** Traces exhibiting severe sensor clipping ($> 2g$) or $\text{SNR} < 3$ dB are unsuitable for spectral analysis; response deconvolution requires valid StationXML.
- **DSP & Numerical Integration:** AC-coupled bandpass integration yields transient dynamic PGD, not permanent tectonic surface ruptures (fling-step).
- **Structural Dynamics & Engineering:** SDOF spectra model single-degree-of-freedom linear elastic oscillators (5% damping), not non-linear multi-story inelastic behavior.

11.3 Cloud Web Access & GitHub Repository

- **Cloud Web Platform:** <https://strong-motion.streamlit.app/> (Zero installation required).
- **GitHub Repository:** <https://github.com/ahmaddidan/BSMA-v.2> (Open-source code, test suite, & docs).

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DEVELOPER PROFILE

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Source Code Repository	: https://github.com/ahmaddidan/BSMA-v.2

[METHODOLOGY CAUTION] ACADEMIC & OPERATIONAL DISCLAIMER

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